Flowrate

README

This notebook serves to provide an overview for estimated values of the:

- mass flow rate
- volume flow rates at different points in the system
- outflow hole radius
- vertical resolution of the system, dependent on time between complete air exchange
- volume flow rate of the vacuum pumps at different altitudes

In order to run the last section of the notebook, you need the file `sodankyla_october_2025_mean_profile.nc` (downloadable from the drive under Thermal) and adjust the `PATH` variable in the `load_pressure_table` function to where you have saved the file.

```
import numpy as np
import matplotlib.pyplot as plt
import xarray as xr
```

Mass flow rate and volumetric flow rate

Definition of mass flow rate and volumetric flow rate

Mass flow rate $\dot{m}$: Defines how much mass is flowing through a surface per unit time.

Volumetric flow rate Q: Defines how much volume is flowing through a surface per unit time.

Because pumps work by pushing a certain amount of volume from the intake side to the outtake side, their flow rate is given as volumetric flow rate. To convert that to a mass flow rate, the volumetric flow rate needs to be multiplied with the gas density.

$$\begin{aligned}\dot{m} &= Q\rho \\ &= Q\frac{p}{R_sT},\end{aligned}$$

where we have used the ideal gas law to determine the density.

Calculation of the mass flow rate and volumetric flow rates of vacuum pumps

The experiment aims to measure atmospheric gas concentrations up to an altitude of 22 km. If used in a parallel configuration, the vacuum (diaphragm) pumps have to work under decreasing levels of intake (ambient) pressures while always providing 1 bar of discharge

(interstage) pressure. According to their data sheet (can be found on the drive under Mechanical > CDR > Data sheet) the vacuum pumps can only provide a volumetric flow rate of around 0.2 l/min under these conditions.

Therefore, we are planning to use the pumps in a series configuration, where one vacuum pumps feeds into the second, which feeds into the compressor. This way both pumps only have to push against a pressure differential between intake and discharge side of 0.5 bar, instead of 1 bar: The first vacuum pump from 0.04 bar to 0.5 bar and the second from 0.5 bar to 1 bar. This means that their volumetric flow rate is around 1 l/min, maybe even a bit higher.

At 22 km altitude, the mass flow rate provided by the vacuum pumps will be the smallest encountered during ascend. We will use this smallest expected mass flow rate to define the radius of the outflow hole of the pressure chamber/pipe 5. As the outflow of the hole is expected to be choked, the outflow speed will be constant at Mach 1 during the whole flight. We can therefore assume a constant mass flow rate out of the pressure chamber during the entire ascend (subject to temperature variations). Therefore, we would like to have the same mass flow rate into the system (vacuum pumps) through the entire ascend as well.

```
R_s = 287 # J/kg/K
p_1 = 0.04 #bar
p_1 *= 100_000 # Pa
T_1 = 273.15 # K, 0°C
rho_1 = p_1/(R_s*T_1) # kg/m^3 = g/l
Q_dia_1 = 1 # l/min
m_flow = rho_1*Q_dia_1 # g/min
print ("Diaphragm pump 1:")
print (f"Inflow temperature: {T_1-273.15:.2f}°C, Inflow pressure:
{p_1/100_000:.2f} bar, Inflow air density: {rho_1:.2f} g/L")
print (f"Mass flow: {m_flow:.2f} g/min")
print (f"Volumetric flow of vacuum pump 1: {Q_dia_1:.2f} l/min\n")


p_2 = 0.5 #bar
p_2 *= 100_000 # Pa
T_2 = 273.15 # K, 0°C
rho_2 = p_2/(R_s*T_2) # kg/m^3 = g/l
Q_dia_2 = m_flow/rho_2 # l/min
print ("Diaphragm pump 2:")
print (f"Inflow temperature: {T_2-273.15:.2f}°C, Inflow pressure:
{p_2/100_000:.2f} bar, Inflow air density: {rho_2:.2f} g/L")
print (f"Mass flow: {m_flow:.2f} g/min")
print (f"Volumetric flow of vacuum pump 2: {Q_dia_2:.2f} l/min")
```

```
Diaphragm pump 1:
Inflow temperature: 0.00°C, Inflow pressure:0.04 bar, Inflow air density:
0.05 g/L
```

```
Mass flow: 0.05 g/min
Volumetric flow of vacuum pump 1: 1.00 l/min

Diaphragm pump 2:
Inflow temperature: 0.00°C, Inflow pressure:0.50 bar, Inflow air density:
0.64 g/L
Mass flow: 0.05 g/min
Volumetric flow of vacuum pump 2: 0.08 l/min
```

The second vacuum pump has to be set to the same mass flow as the first vacuum pump. This is done in order to keep the pressure between the vacuum pumps at 0.5 bar. Similarly, the compressor needs to be set to use the same mass flow rate to keep the pressure between the second vacuum pump and itself at 1 bar.

However, a constant mass flow rate at different gas pressures and temperatures necessitates varying volumetric flow rates.

Calculation of volumetric flow rate of the compressor

```
R_s = 287 # J/kg/K
p_3 = 1 #bar
p_3 *= 100_000 # Pa
T_3 = 273.15 +10 # K, 10°C
rho_3 = p_3/(R_s*T_3) # kg/m^3 = g/l
Q_comp = m_flow/rho_3 #l/min
print ("Compressor:")
print (f"Inflow temperature: {T_3-273.15:.2f}°C, Inflow pressure:
{p_3/100_000:.2f} bar, Inflow air density: {rho_3:.2f} g/L")
print (f"Volumetric flow: {Q_comp:.2f} l/min")
```

```
Compressor:
Inflow temperature: 10.00°C, Inflow pressure:1.00 bar, Inflow air density:
1.23 g/L
Volumetric flow: 0.04 l/min
```

Hole Size for Outflow

The mass going into the chamber has to be equal to the mass leaving the chamber. The mass flow into the chamber is the same as the one through the pumps. The mass flow out of the chamber can be constructed from volumetric flow and air density:

$$
\begin{aligned}
\dot{m}_{in} &= \dot{m}_{out} \\
&= Q_h \rho_{ch} \\
&= \frac{V_h}{t} \rho_{ch} \\
&= A_h v \rho_{ch} \\
&= \pi r^2 v \rho_{ch} \\
\Rightarrow r &= \sqrt{\frac{\dot{m}_{in}}{\pi v \rho_{ch}}}
\end{aligned}
$$

with v the flow speed (sound velocity), $A_h = \pi r^2$ the area of the hole, and ρ_{ch} the chamber air density

```
v = 343 #m/s, for 20°C; at -40°C it's only 306m/s
v = v*10*60 #dm/min

p_4 = 3 #bar
p_4 *= 100_000 # Pa
T_4 = 273.15 +20 # K, 10°C
rho_4 = p_4/(R_s*T_4) # kg/m^3 = g/l

r = np.sqrt(m_flow/(np.pi*v*rho_4)) # dm

print(f'Radius of Outlet hole: r={r*100:.4f}mm')
```

```
Radius of Outlet hole: r=0.0149mm
```

Varying the volumetric inflow of the vacuum pumps to see changes in spatial resolution and hole size

Because the mass flow in and out of the chamber is quite low in comparison with the total mass inside the chamber, at any given altitude a significant part of the chamber volume will be filled with air collected at lower altitudes. For the scientific evaluation of the collected it is important to know how quickly the air is completely exchanged inside the pressure chamber. In this section we will vary the volumetric flow rate of the first vacuum pump, to see how higher or lower flow rates affect the hole size of the pressure chamber as well as the vertical distance travelled between complete air exchanges

The time to complete one whole air cycle is calculated by the cycle_time function below. For a derivation see the "Fill and flow rates MIRAGE" document on the drive under Science. The volumetric flow rate of the chamber can be calculated from

$$Q_{ch} = \dot{m}/\rho_{ch}$$

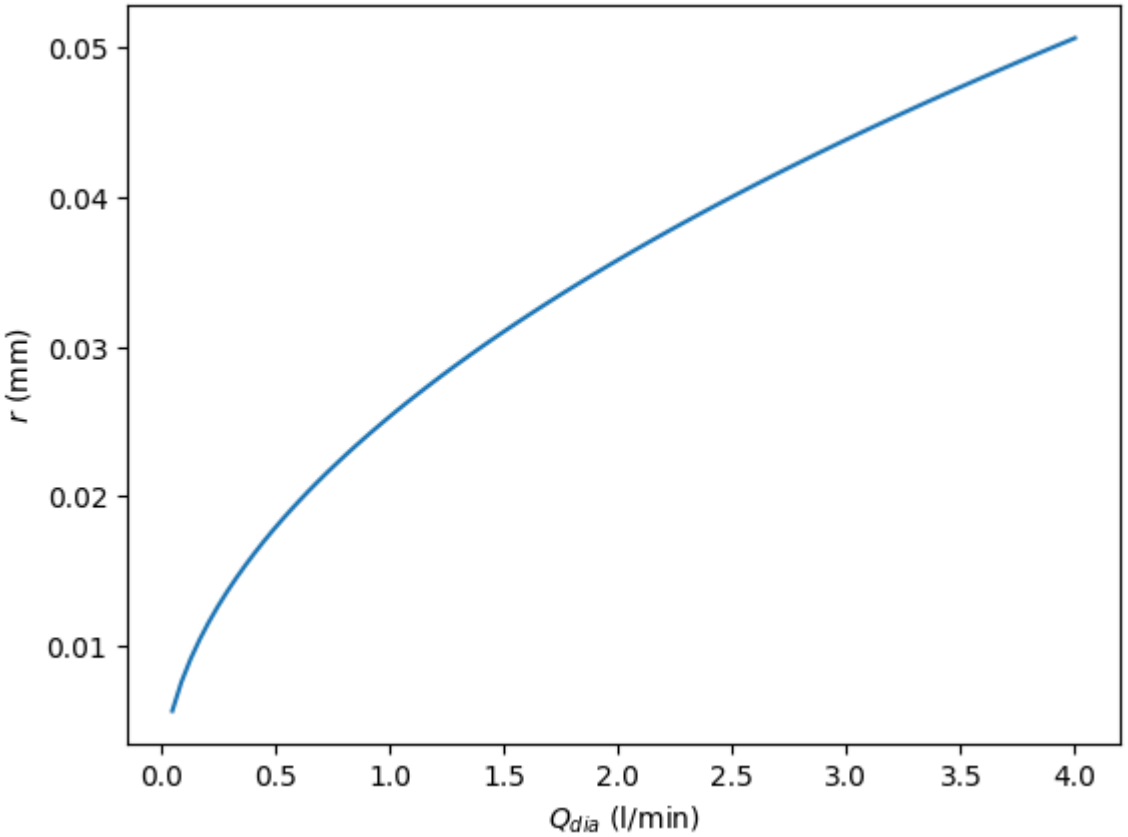
```
def cycle_time(Q, V_l=0.5):
```

```
    t_s = -np.log(1 - 0.997) * V_l / Q
    return t_s

ascend_speed = 5 # m/s
```

```
r_arr = []
Delta_h_arr = []

rho_1 = p_1/(R_s*T_1)
rho_2 = p_2/(R_s*T_2)


Qin_array = np.linspace(0.05, 4, 100)
for q in Qin_array:
    m_flow_changed = rho_1*q
    r = np.sqrt(m_flow_changed/(np.pi*v*rho_3))*100 # mm
    r_arr.append(r)

    Q_chamber = m_flow_changed/rho_3
    t_s = cycle_time(Q_chamber, V_l = 0.045)
    Delta_h = t_s*ascend_speed

    Delta_h_arr.append(Delta_h)
```

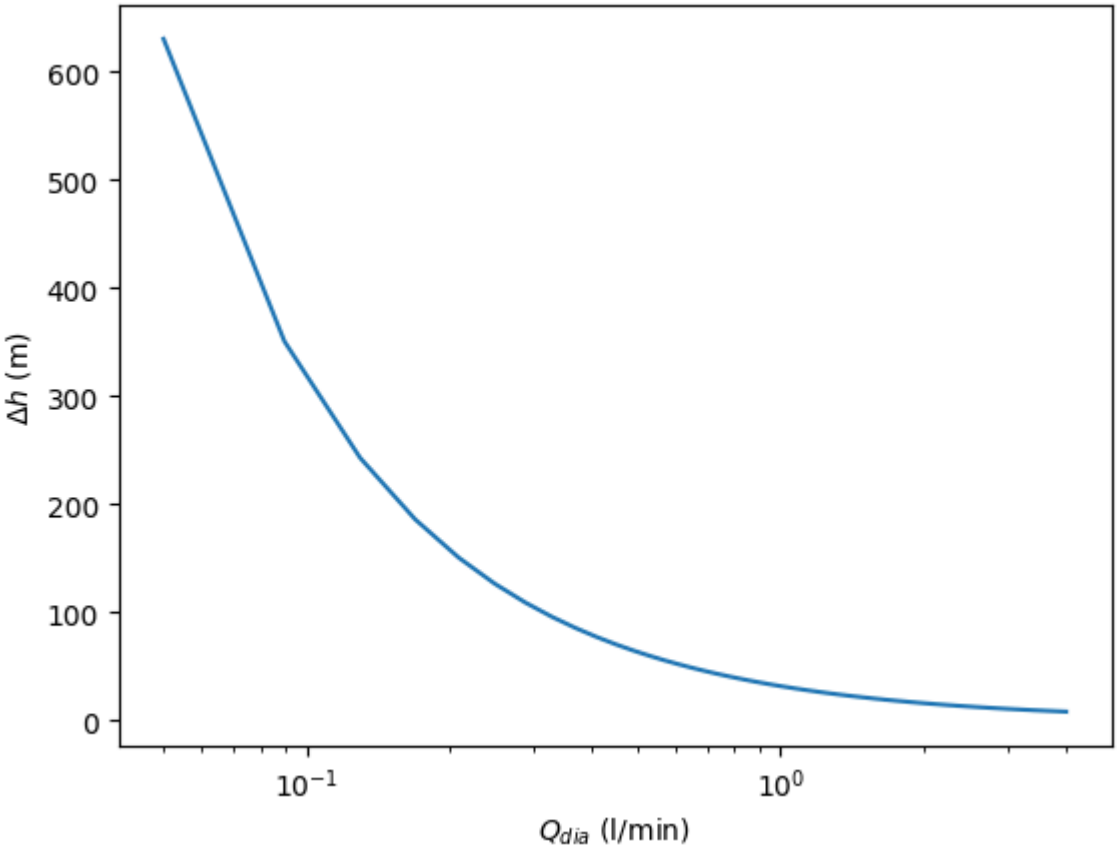
```
plt.xlabel(r'$Q_{dia}$ (l/min)')
plt.ylabel(r'$r$ (mm)')
plt.plot(Qin_array, r_arr)
```

```
[<matplotlib.lines.Line2D at 0x110220100>]
```




```
plt.xlabel(r'$Q_{dia}$ (l/min)')
plt.ylabel(r'$\Delta h$ (m)')
#plt.yscale('log')
plt.xscale('log')
plt.plot(Qin_array, Delta_h_arr)
```

```
[<matplotlib.lines.Line2D at 0x110287970>]
```



How does the required volumetric flow rate of the pumps change with ambient pressure?

Since the mass flow is always constant and the pressure in the pipe between compressor and pump is always constant at 1 bar, the compressor always has a constant volumetric flow rate as well.

Let's look at the first vacuum pump. The volumetric flow rate is defined as

$$\begin{aligned} Q &= \frac{\dot{m}}{\rho_{ambient}} \\ &= \dot{m}\frac{p_{ambient}}{R_s * T_{ambient}} \end{aligned}$$

Let's have a look at what that means using the averaged temperature and pressure profile obtained from the Sodankyla weather station.

```
def load_pressure_table(
    filename="sodankyla_october_2025_mean_profile.nc", return_ds=False
):
    """
        Reads the pressure table and returns arrays:

    heights (m)
    pressures (hPa)
    temperatures (K)
    """
```

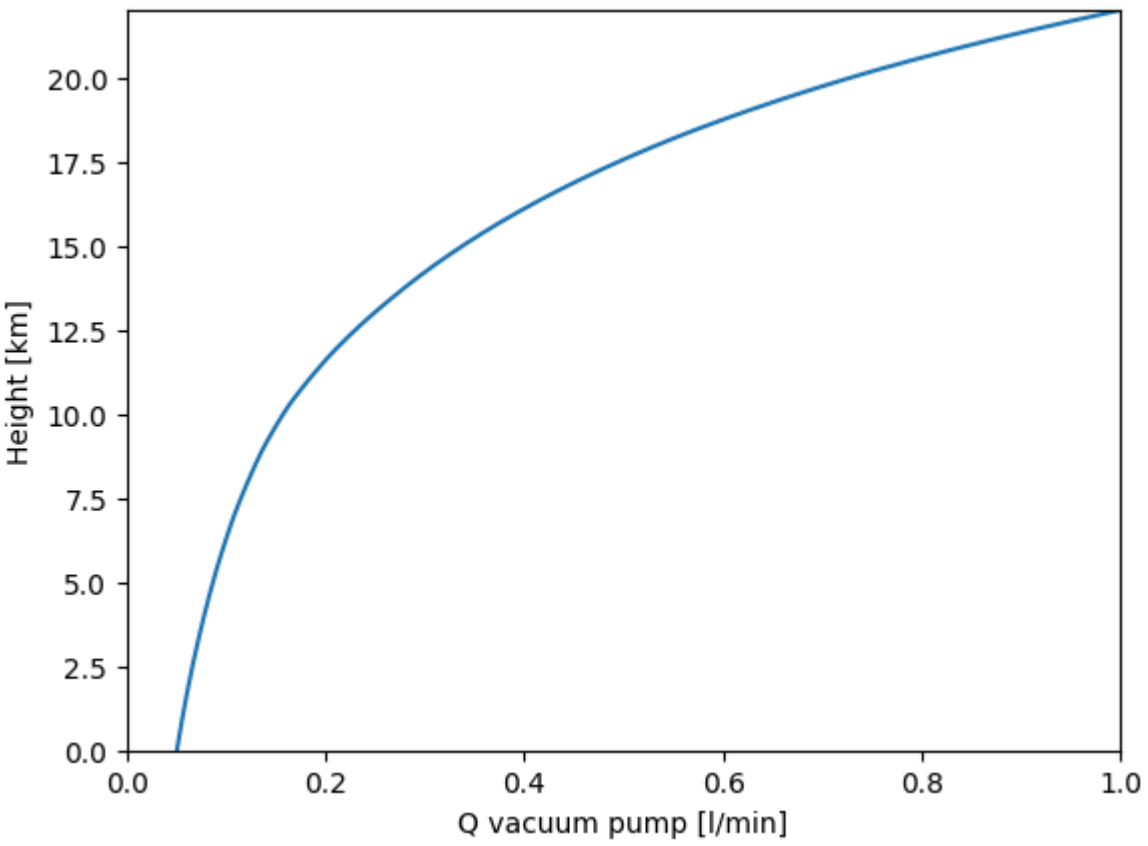
```
    PATH = "~/Documents/RexusBexus/code/python/analysis/"
    filename = PATH+filename

    if filename.split(".")[-1] == "nc":
        # we can load the file as an xarray
        ds = xr.open_dataset(filename)
        if return_ds:
            return ds
        else:
            return ds["height"], ds["pressure_mean"],
ds["temperature_mean"] + 273.15

ds = load_pressure_table(return_ds=True)
```

Now let's calculate the mass flow rate expected at maximum altitude

```
R_s = 287  # J/kg/K

Q_vac_max = 1 # l/min

# Defining the mass flow rate again at high altitudes
z_max = 22*1000 #m
ds_subset = ds.sel(height=z_max, method='nearest')
T_max = ds_subset['temperature_mean'] # °C
p_max = ds_subset['pressure_mean'] # hPa

rho_max = p_max*100/(R_s*(T_max+273.15)) # kg/m^3 = g/l

mdot_max = Q_vac_max*rho_max # g/min

print (f'Mass flow rate at maximum altitude: {mdot_max:.2f} g/min')
```

```
Mass flow rate at maximum altitude: 0.06 g/min
```

Given the mass flow rate at maximum altitude, we can calculate the volume flow rate of the first vacuum pump for different altitudes

```
rho_ambient = ds['pressure_mean']*100/(R_s*
(ds['temperature_mean']+273.15)) # kg/m^3 = g/l
Q_vac = mdot_max/rho_ambient # l/min

plt.plot(Q_vac, ds['height']/1000)
plt.ylim(0,z_max/1000)
plt.xlim(0,Q_vac_max)
plt.ylabel('Height [km]')
plt.xlabel('Q vacuum pump [l/min]')
```

```
Text(0.5, 0, 'Q vacuum pump [l/min]')
```